

Automated and Interpretable Grading for Embryo Fragmentation via Deep Learning

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Abstract—Embryo fragmentation grade serves as a pivotal morphological indicator for assessing developmental potential in In Vitro Fertilization (IVF). However, despite its clinical significance, current deep learning approaches often suffer from a lack of interpretability and continue to rely on manual post-processing. This dependence on human intervention reintroduces subjectivity into the grading process, undermining the goal of a fully objective assessment. In this study, we investigate the feasibility of end-to-end automation for this task. We first evaluate a single-task learning (STL) paradigm optimized for pixel-level segmentation of fragmented cells. Fragmentation grades in STL are derived by applying a prescribed formula to the resulting segmentation masks. STL achieved a Dice coefficient of 0.744 and a Mean Absolute Error (MAE) of 0.040, serving as the benchmark in this study due to the optimal segmentation. To address the demand for a fully integrated and interpretable AI solution, we propose a multi-task learning (MTL) framework. In this architecture, a grade-prediction head shares visual features with the segmentation task during training. To optimize this shared learning process, we implemented a decoupled strategy to mitigate gradient interference between the spatial segmentation task and the prediction head. While this MTL approach offers a more streamlined, end-to-end pipeline, it entails an intrinsic performance trade-off. Specifically, the joint optimization strategy yields a precise grading (MAE: 0.046) with slightly smooth boundaries (Dice: 0.716), whereas the decoupled strategy perfectly preserves spatial fidelity (Dice: 0.729) with robust grading (MAE: 0.049). Nevertheless, both configurations confirm the high potential of MTL as a viable, fully automated solution for clinical embryo assessment.

Keywords: Embryo Grading, Multi-Task Learning, Explainable AI, Gradient Conflict, Semantic Segmentation, Decoupled Training.

I. INTRODUCTION

A. Background

Infertility is a critical global public health issue, with assisted reproductive technology (ART)—specifically in vitro fertilization (IVF), serving as the primary clinical intervention. IVF success depends heavily on the embryo transfer outcome, which is largely dictated by embryo morphological quality [1]. Among key morphological indicators, embryo fragmentation, the presence of anucleate cytoplasmic fragments during early

cleavage, is a major negative prognostic factor that impairs developmental potential and reduces implantation rates [2]. Consequently, precise quantitative and qualitative assessment of fragmentation is vital for optimizing embryo selection and improving ART efficacy.

While clinical practice standardly utilizes the Gardner grading system (or similar frameworks) to categorize fragmentation into discrete tiers, Grade A (<10%), Grade B (10–25%), Grade C (25–50%), and Grade D (>50%), this manual evaluation is inherently subjective. Significant inter-observer variability exists even among experienced embryologists [3], [4]. This subjectivity introduces risks of misclassification for viable embryos, ultimately compromising clinical decision-making and patient outcomes.

B. Motivation and Challenges

To mitigate the limitations of manual assessment, Computer-Aided Diagnosis (CAD) systems have been developed. Nevertheless, existing solutions face significant bottlenecks. First, early texture analysis (e.g., GLCM) lacks discriminability for complex embryonic cytoplasmic structures. Second, standard deep learning regression models suffer from the "black-box" issue, failing to provide the visual fragment localization required in medical AI.

An ideal interpretable system must simultaneously output a quantitative grade and a segmentation mask. However, these tasks present an inherent optimization conflict: segmentation prioritizes pixel-level boundary fidelity, whereas grading focuses on global area estimation. Under standard joint optimization, this divergence frequently induces negative transfer, compromising task convergence.

To address these challenges, we propose a dual-paradigm exploration. We first establish a precision benchmark using a Single-Task Learning (STL) approach paired with a deterministic physical conversion formula. In parallel, we develop an attention-guided Multi-Task Learning (MTL) framework to provide a streamlined, end-to-end clinical solution that eliminates the need for manual post-processing.

C. Contributions

The primary contributions of this work are threefold:

1) *Single-Task (Segmentation)*: We propose an STL architecture dedicated to high-fidelity boundary delineation. By replacing the neural network regression with an explicit physical area conversion formula, we demonstrate that when

segmentation is sufficiently accurate (Dice: 0.744), the deterministic derivation yields a superior MAE (0.040), establishing a theoretical accuracy upper bound without relying on a regression head.

2) *Optimization of the Multi-Task Paradigm*: For an end-to-end automated grading solution, we introduce an MTL framework. Crucially, this architecture establishes a strong task synergy, leveraging the high-fidelity spatial representations learned from the segmentation task as a powerful visual prior to enhance the global regression accuracy. To further mitigate inherent gradient interference, we develop a decoupled training strategy equipped with a feature-injection mechanism, which systematically corrects area estimation biases.

3) *Utilization of Weakly Labeled Data*: To address the limited pixel-level annotations in the MTL framework, we incorporate a range-based loss function that converts discrete clinical grades into continuous interval constraints, allowing extensive weakly labeled data to be effectively exploited.

II. RELATED WORK

This section reviews related techniques in embryo grading and, through preliminary experimental data, analyzes the limitations of traditional methods and existing deep learning strategies. This analysis establishes the necessity of our proposed dual-paradigm exploration: first, introducing a single-task learning (STL) approach to achieve high-fidelity segmentation without gradient conflicts; and second, developing a multi-task learning (MTL) framework to provide a fully automated, end-to-end grading solution.

A. Limitations of Hand-crafted Features

Historically, texture analysis served as the primary method for quantifying embryo quality. Given that fragmentation primarily manifests as textural anomalies disrupting cytoplasmic uniformity, it is essential to first evaluate the efficacy of classic descriptors, such as the Gray-Level Co-occurrence Matrix (GLCM), to justify the transition to deep learning architectures.

1) *Data Selection and Clinical Relevance*: To ensure the high clinical relevance of our experimental results, all preliminary experiments in this section specifically focus on the Early Cleavage Stage, using embryo images at the T2 (2-cell) and T4 (4-cell) stages. This selection is based on two reasons: First, literature indicates that the degree of fragmentation during the cleavage stage (Day 2-3) is a key early biomarker for predicting whether an embryo can successfully develop to the blastocyst stage [7], [8]. Second, compared to compacted morulae, the cell boundaries and fragment structures at the T2/T4 stages are relatively distinct, making them an ideal scenario for verifying whether algorithms can effectively distinguish between cytoplasmic noise and true fragments.

2) *GLCM Analysis*: Based on the T2/T4 dataset, we extracted GLCM texture features from the region of interest (ROI) of embryos and visualized them using t-SNE dimensionality reduction. As shown in Fig. 1, even samples with the most disparate grades (Grade A vs. Grade D) exhibit severe overlap in the feature space. This indicates that GLCM features lack sufficient discriminability to overcome common background noise in T2/T4 embryo images, such as

cytoplasmic granules and vacuoles, failing to form clear decision boundaries.

3) *Failure of Non-linear Mapping*: Given the inadequacy of GLCM, we extended our evaluation to Local Binary Patterns (LBP) [9] features, which are more sensitive to local details, and trained a Multi-Layer Perceptron (MLP) for non-linear regression. Although the model could identify samples with extreme fragmentation (Grade D), it failed to discriminate the subtle differences between intermediate grades (Grade A, B, and C), as illustrated in Fig. 2. This experiment confirms the fundamental deficiency of hand-crafted features in capturing subtle morphological changes in early embryos, showcasing the necessity of shifting to automated feature learning via deep neural networks.

B. Deep Learning Baselines

1) *Regression Models (The Black-Box Issue)*: We employed standard CNN architectures (e.g., InceptionV3) to train regression models for directly predicting fragmentation ratios. As shown in Fig. 3, the distribution of predicted values aligns well with the ground truth trends. However, this approach suffers from a fundamental clinical flaw: the black-box nature of deep learning. The model outputs only a numerical score without providing any visual evidence of fragment locations. This absence of visual verification prevents clinicians from validating the basis of the grading, thus failing to meet the strict explainability requirements in medical AI

2) Role of Semantic Segmentation

To address the black-box issue, an alternative approach employs semantic segmentation (e.g., standard DeepLabV3+ [10]) to extract fragment regions for indirect grading. Although this provides excellent explainability, our

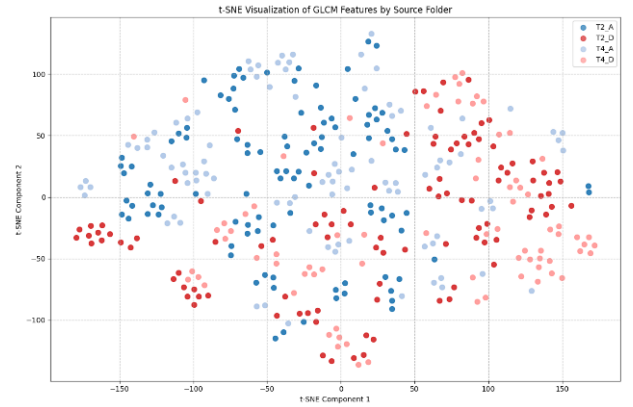


Fig. 1. t-SNE visualization of GLCM texture features.

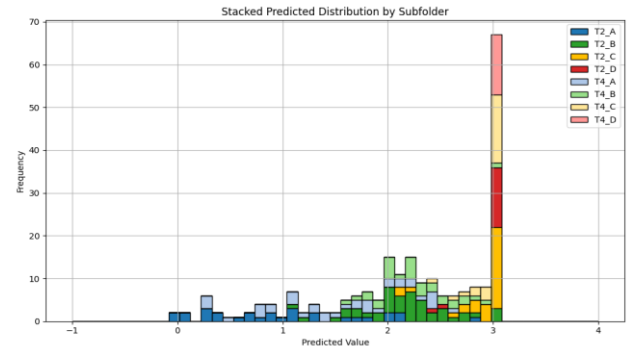


Fig. 2. Prediction distribution of LBP-based MLP model.

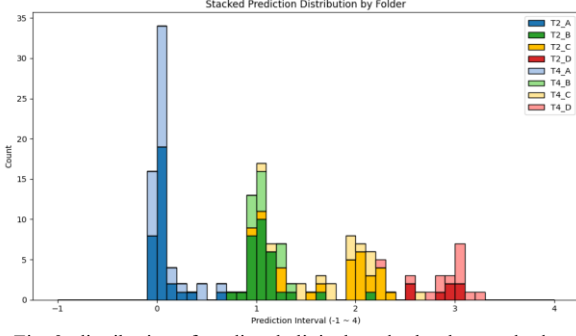


Fig. 3. distribution of predicted clinical grades by the standard regression baseline (inceptionv3).

experiments revealed that the Mean Absolute Error (MAE) derived from standard segmentation masks is often unsatisfactory. Due to blurred and irregular embryo boundaries, minor pixel-level errors in conventional models accumulate into significant area deviations. This suggests that without rigorous boundary optimization, standard segmentation architectures are insufficient for direct geometric derivation.

C. Shape-area Trade-off

Ideally, a system should concurrently yield precise numerical evaluations and fine-grained segmentation masks. However, integrating these tasks introduces an inherent structural tension: segmentation prioritizes high-frequency boundary details, whereas regression emphasizes low-frequency global abstractions. This competing relationship often induces Gradient Conflict and Negative Transfer [5], [6] in a shared backbone. To fundamentally resolve this "shape-area trade-off" and meet diverse clinical needs, this study explores and compares two distinct paradigms:

1) *Establishing the Precision Baseline (STL)*: We first evaluate the network's capacity when dedicated entirely to boundary preservation. We hypothesize that the resulting high-fidelity mask enables highly accurate, deterministic physical area conversion, serving as a rigorous benchmark for subsequent multi-task extensions.

2) *Mitigating Interference (MTL)*: Recognizing the demand for fully automated, end-to-end pipelines, we develop a decoupled MTL [11] strategy equipped with a feature-injection mechanism. This harmonizes divergent gradients to provide a streamlined alternative, albeit with a slight performance degradation.

D. Continuous Handling of Discrete Grades

Clinically, grades A, B, C, and D represent distinct intervals within a continuous proportional scale (0–100%) [12]. Treating the grading task as a standard one-hot encoding problem would discard critical magnitude and ordinal information. To address this, we reformulate the grading task as a continuous ratio regression problem and utilize discrete labels to establish explicit interval constraints. This approach is analogous to the granular evaluation strategy (i.e., first segmentation, then classification) proposed by [13]. Our approach aligns more closely with clinical reality and ensures stable model convergence.

III. METHODOLOGIES

This section details our dual-paradigm exploration for embryo fragmentation grading. We first introduce a high-fidelity Single-Task Learning (STL) framework designed for precise boundary delineation. Subsequently, we present an extended Multi-Task Learning (MTL) framework that incorporates a regression branch to achieve a fully automated, end-to-end grading solution. Crucially, within the MTL framework, we formalize two distinct optimization strategies—Full MTL under joint optimization and a Two-Stage Decoupled strategy—to systematically address the inherent gradient conflict.

A. STL Framework

In Fig. 4, the STL framework is designed to achieve high-fidelity boundary delineation. STL utilizes a standard ResNet-50 [14] to form a multi-scale feature pyramid, incorporating an Atrous Spatial Pyramid Pooling (ASPP) module at the deepest bottleneck (f_4 , $1/32$ scale). During the top-down decoding phase, Attention Gates (AGs) [15] are integrated into the U-Net skip [16] connections. To recover fine details without introducing cytoplasmic noise, AGs replace standard skip connections to actively filter shallow features before deep feature concatenation. Let F_1 denote the shallow features and g be the upsampled deep gating signal. The attention coefficient map α is formulated as

$$\alpha = \sigma \left(\psi \left(\text{ReLU}(W_x(F_1) + W_g(g) + b) \right) \right). \quad (1)$$

This hierarchical attention mechanism leverages deep semantic context to filter out background artifacts and cytoplasmic noise at multiple scales. By dedicating the network's capacity exclusively to the boundary objective ($L_{\text{seg}} + L_{\text{bound}}$), the output feature map achieves precise micro-fragment contour preservation.

B. Conversion Formula

Based on the high-fidelity binary mask (M_{frag}) generated by the STL framework, the fragmentation grade is assessed through a transparent, deterministic geometric mapping. The fragmentation ratio is deterministically derived using Eq. (2):

$$\text{Frag} = \frac{\sum_{i,j} M_{\text{frag}}(i,j)}{\sum_{i,j} M_{\text{total}}(i,j)}. \quad (2)$$

where M_{total} represents the total pixel area of the embryo. This physical derivation provides visual and clinical explainability. Furthermore, we posit that with sufficiently precise spatial segmentation boundaries (high Dice), this explicit

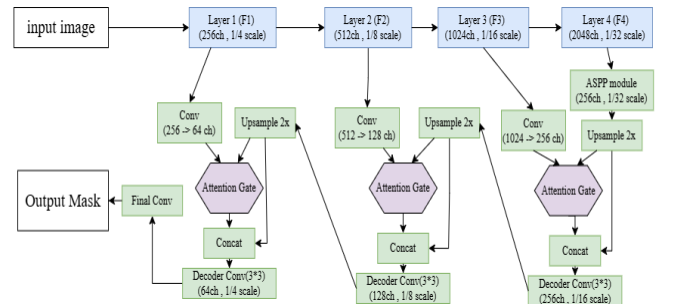


Fig. 4. Architecture of STL comprising a multi-scale encoder (Layers 1-4) and an ASPP module to capture global context. Attention Gates are integrated into the skip connections to systematically filter cytoplasmic noise from shallow features using deep semantic gating signals.

mathematical derivation inherently outperforms black-box neural network regression in Mean Absolute Error (MAE).

C. Paradigm II: Multi-Task Learning

To meet the demand for end-to-end automation, we extend the STL architecture into an MTL framework. As illustrated in Fig. 5, this paradigm incorporates a parallel regression branch, integrating a shared backbone with two task-specific modules: one for global quantification and another for local boundary delineation.

1) *Shared Backbone*: We utilize ResNet-50 as the shared feature extractor. To accommodate segmentation resolution requirements, we remove the downsampling strides in Layers 3 and 4, employing dilated convolutions (rates 2 and 4) to maintain an output stride of 8 for the final feature map (F_4). This preserves semantic abstraction without sacrificing spatial details. The model extracts low-level F_1 features for boundary recovery, and high-level F_3 and F_4 features as core inputs for context and regression. By jointly optimizing with the segmentation branch, the backbone is explicitly regularized to capture high-frequency spatial boundaries. This induces a powerful task synergy, providing a highly discriminative feature foundation for the regression module.

2) *Regression Module*: To map images to a continuous grading scale $\hat{y}_{\text{reg}} \in [0, 1]$, we design a multi-scale regression head. Fused features from F_3 and F_4 are processed via Global Average Pooling (GAP) and a Multi-Layer Perceptron (MLP). A latent feature vector $V_{\text{reg}} \in \mathbb{R}^{2048}$ is extracted from the penultimate MLP layer. Representing global fragmentation severity, this vector is injected into the segmentation branch to guide pixel-level predictions.

3) *Feature-Injected Segmentation Module*: Inheriting the ASPP and Attention Gates from the STL framework, the MTL segmentation decoder integrates one additional crucial component to correct systematic area estimation errors: Regression-Guided Feature Injection. To align local segmentation with the global severity grade, the regression vector V_{reg} is spatially broadcast and fused with the decoder features F_{dec} . By conditioning the final convolution on this

fused representation (F_{final}), the model seamlessly incorporates global priors to rectify local ambiguities.

D. Hybrid Loss and Optimization Strategies

To coordinate the pixel-level and image-level tasks under semi-supervised settings, we minimize a composite objective:

$$L_{\text{total}} = \alpha L_{\text{seg}} + \beta L_{\text{reg}} + \gamma L_{\text{cons}}. \quad (3)$$

1) *Hybrid Loss Components*: L_{seg} is defined as the weighted sum of Binary Cross Entropy, Dice Loss [17], and Focal Loss [18] to address class imbalance. For the regression branch, we employ a precise L_1 loss for paired data. For weakly labeled data (clinical grades only), we propose a continuous Range Loss (L_{range}), penalizing predictions only when they violate interval bounds derived from discrete grades.

$$L_{\text{range}}(\hat{y}, G) = \text{ReLU}(y_{\min} - \hat{y}) + \text{ReLU}(\hat{y} - y_{\max}). \quad (4)$$

Finally, a Consistency Loss (L_{cons}) minimizes the discrepancy between the regression prediction and the mask-derived area.

2) *Multi-Task Optimization*: To evaluate the impact of gradient dynamics and task interference, we formalize two distinct training regimes utilizing the composite objective:

Full MTL Strategy (Joint Optimization): In this configuration, the entire network parameters—including the shared backbone θ_{bckbn} , the segmentation decoder θ_{seg} , and the regression head θ_{reg} —are optimized simultaneously in an end-to-end manner by minimizing L_{total} with all balancing weights active ($\alpha, \beta, \gamma > 0$). While this joint optimization provides a streamlined pipeline, the shared feature space remains highly susceptible to Negative Transfer due to competing gradient vectors between the divergent targets.

Two-Stage Decoupled Strategy: To actively mitigate negative transfer, we propose a sequentially decoupled alternative. Phase 1 (Visual Expert Pre-training): The shared backbone and decoder are optimized using only L_{seg} and L_{cons} by setting $\beta = 0$. This restricts the network capacity to capturing high-frequency spatial attributes, establishing optimal visual fidelity. Phase 2 (Regression-guided

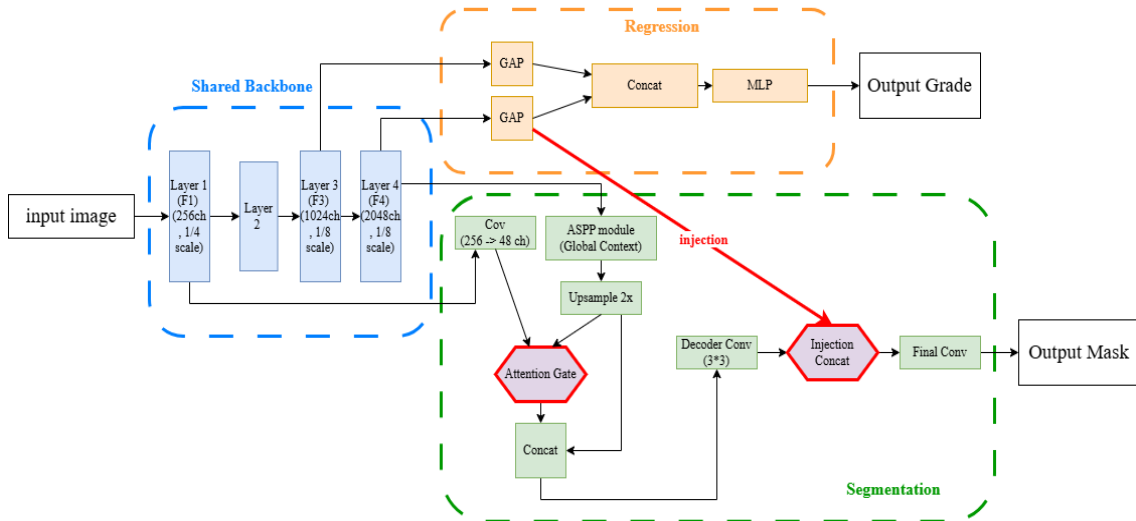


Fig. 5. The framework of MT, comprising a shared ResNet-50 backbone, a segmentation branch, and a regression branch. The red arrow indicates the U-Net skip connection.

finetuning): To address the gradient conflict, the learned parameters of the shared backbone are frozen:

$$\theta_{\text{bckbn}}^{(t+1)} = \theta_{\text{bckbn}}^{(t)} \quad (\text{Frozen}). \quad (5)$$

Optimization is restricted to the regression head θ_{reg} and the feature injection layers, explicitly preventing low-frequency regression gradients from disrupting the spatial boundaries.

IV. EXPERIMENTAL RESULTS

To verify our dual-paradigm exploration, rigorous ablation studies were conducted. We evaluate the models to first establish a precision baseline achieved without gradient interference, and subsequently quantify the performance trade-off associated with achieving a fully automated, end-to-end grading solution.

A. Experimental Setup

1) *Data partitioning*: This study utilizes clinical embryo images focused on the critical cleavage stage (2-cell and 4-cell). To evaluate the efficacy of semi-supervised learning, the dataset was split into two subsets. First, Paired Data (D_{paired}) with 373 samples annotated with both pixel-level fragmentation masks and clinical grades. This subset supports fully supervised training for segmentation (L_{seg}) and precise regression (L_{pre}). Second, Grade-Only Data (D_{weak}) of 1549 samples annotated with clinical grades without segmentation masks. This subset is used exclusively for the weakly supervised regression via Eq.(4).

2) *Preprocessing*: To prevent geometric distortion, input images were resized to 299×299 pixels using an aspect-ratio-preserving zero-padding strategy.

3) *Implementation Details*: The framework was implemented in PyTorch using the AdamW [19] optimizer. To ensure training stability across different configurations, distinct learning rate schedules were employed:

For the Single-Task and Full MTL strategy (Baseline, Exp A to D, and F): A consistent base learning rate of 1×10^{-4} was utilized.

For the Decoupled MTL strategy (Exp E): A differential learning rate schedule was applied, where a base learning rate of 1×10^{-4} was used during Phase 1 (Visual Expert Pre-training), and a strictly reduced rate of 1×10^{-5} was applied during Phase 2 (Regression-Guided Finetuning) to optimize the regression head while preserving the pre-trained spatial features.

4) *Evaluation Metrics*: Performance was assessed using the Dice Similarity Coefficient (Dice) for spatial overlap (shape fidelity) and MAE for grading accuracy.

B. Architecture Selection and Verification

Before investigating multi-task strategies, we validated the proposed attention-guided segmentation backbone. Compared to the standard DeepLabV3+ [10] baseline (Dice: 0.717)—which tends to over-smooth and misclassify cytoplasmic granules—our attention-guided architecture (Exp B) successfully elevates segmentation accuracy to 0.729. This confirms that filtering high-frequency noise via Attention Gates improves boundary discriminability, establishing Exp B as a robust visual baseline for subsequent multi-task extensions.

C. Ablation Study

Ablation study was conducted to systematically quantify the gradient trade-off in multi-task learning.

1) *Experiment settings*: To isolate the contribution of each component, we defined seven configurations alongside the Standard DeepLabV3+ Baseline (see Table 1):

Exp A (STL): The proposed foundational single-task architecture. It derives fragmentation grades directly through deterministic physical area conversion based on high-fidelity segmentation masks, establishing the absolute precision baseline.

Exp B (Visual Baseline / Phase 1 Pre-training): Utilizes the segmentation branch of the MTL framework independently ($L_{\text{seg}} + L_{\text{cons}}$). It establishes the segmentation upper bound without the gradient interference from the regression task.

Exp C (Regression Baselines): Regression-only models to benchmark quantification without spatial supervision. Stratified into Exp C1 (Paired Only) and Exp C2 (Semi-supervised with D_{weak}).

Exp D (Full MTL): The complete MTL with feature injection under joint optimization, optimizing L_{total} jointly.

Exp E (Decoupled Strategy): Adopts the pretrain-freeze-finetune strategy to verify positive transfer without catastrophic forgetting.

Exp F (No-Inject Control): An ablation for Exp D where the Feature Injection mechanism is disconnected.

2) *Quantitative Analysis*: Table 1 validates our hypotheses regarding feature interaction and gradient dynamics.

Analysis of the Precision Ceiling (Exp A): As established, utilizing the attention-guided segmentation branch of the MTL framework independently yields a robust visual baseline (Exp B, Dice: 0.729). In contrast, the dedicated STL architecture (Exp A) establishes the absolute precision ceiling for this study. By exclusively optimizing the network for boundary preservation without any multi-task structural constraints, Exp A achieves an optimal Dice score of 0.744. When applying the explicit physical conversion formula to this high-fidelity mask, the MAE drops to 0.040. This confirms that when spatial fidelity is maximized, deterministic physical derivation provides the most accurate and interpretable grading.

Table 1: Quantitative Results of Ablation Study

Exp ID	Architecture	Strategy	Loss Setting	Dice Score	MAE
Baseline	DeepLabV3+	Single Task	$L_{\text{seg}} + L_{\text{cons}}$	0.717	N/A
Exp A	STL+ Eq. (2)	Single Task	$L_{\text{seg}} + L_{\text{bound}}$	0.744	0.040
Exp B	MTL (Seg)	Single Task	$L_{\text{seg}} + L_{\text{cons}}$	0.729	N/A
Exp C1	Regression	Single Task	L_{pre}	N/A	0.057
Exp C2	Regression	Single Task	$L_{\text{reg}} (L_{\text{pre}} + L_{\text{range}})$	N/A	0.051
Exp D	MTL	Full MTL	L_{total}	0.716	0.046
Exp E	MTL	Decoupled	Phase 1: L_{seg} Phase 2: L_{reg}	0.729	0.049
Exp F	MTL (No-Inject)	Full MTL	L_{total}	0.678	0.053

Validation of Feature Injection (Exp D vs. Exp F):

Ablating the injection mechanism under joint training (Exp F) resulted in significant performance degradation (Dice: 0.678, MAE: 0.053). This demonstrates that naively summing losses induces severe gradient conflict. The feature injection mechanism is essential to coordinate divergent tasks, providing global semantic priors to the segmentation decoder.

Efficacy of Weak Labels (Exp C1 vs. Exp C2): Training solely on limited paired data yielded an MAE of 0.057. Incorporating grade-only data via interval constraints decreased the error to 0.051. This confirms that L_{range} successfully extracts global priors from discrete grades, mitigating data scarcity.

Task Synergy and Gradient Trade-off (Exp C, D, & E):

Full MTL (Exp D, MAE: 0.046) outperforms the pure regression baseline (Exp C2, MAE: 0.051), demonstrating strong task synergy. The auxiliary segmentation acts as a robust visual prior, capturing high-frequency boundaries to improve global regression. However, Exp D exhibits a gradient trade-off, slightly sacrificing boundary details (Dice: 0.716) compared to the Exp B visual baseline (0.729). Alternatively, Exp E (Decoupled) perfectly preserves SOTA segmentation (Dice: 0.729) with robust regression (MAE: 0.049), balancing explainability and stability.

Summary: Ultimately, STL (Exp A) proves that high spatial fidelity combined with deterministic physical derivation yields the most accurate and transparent solution. Conversely, for clinical workflows prioritizing end-to-end automation, MTL configurations offer highly robust alternatives. By leveraging spatial features as visual priors, they effectively balance precision and operational efficiency.

V. CONCLUSION

Our key findings include:

(1) The STL Precision Baseline: Dedicated entirely to boundary preservation, the STL framework (Exp A) achieves superior spatial fidelity (Dice: 0.744). This enables deterministic area calculation with minimal error (MAE: 0.040), establishing the theoretical accuracy benchmark. (2) MTL and Task Synergy: For end-to-end clinical automation, we extended the architecture into an MTL framework (Exp D). Compared to regression-only baselines (Exp C), spatial features from the auxiliary segmentation task act as a powerful visual prior, significantly improving regression accuracy (MAE: 0.051 to 0.046). Although joint optimization induces a slight gradient trade-off (Dice: 0.716), our Feature Injection mechanism effectively coordinates these divergent tasks, preventing the catastrophic degradation observed in its absence (Exp F).

Ultimately, STL provides the most transparent solution for absolute precision, whereas MTL offers a fully automated alternative balancing accuracy and operational efficiency. Future research will integrate time-lapse data and additional morphological features for comprehensive spatiotemporal scoring. To achieve real-world clinical translation, we will next evaluate deployment on resource-constrained edge

systems, aligning with recent edge-AI methodologies that deploy end-to-end deep fusion [20] on restricted devices.

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